

TrES-3b

R-2622

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_220605 · created 2026-08-28T23:35:08+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459735.8963 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.014
Transit depth (Rp/Rs)^2	2.4 +/- 0.43 [%]
Orbital Inclination (inc)	85.67 +/- 0.79
Airmass coefficient 1 (a1)	1.05 +/- 0.015
Airmass coefficient 2 (a2)	-0.042 +/- 0.029
Scatter in the residuals of the lightcurve fit is	1.43 %
Best Comparison Star	#2 - [303, 310]
Optimal Aperture	3.73
Optimal Annulus	11.1
Transit Duration (day)	0.0766 +/- 0.003

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.155 ± 0.014 vs 0.16309 (deviation 0.54σ)
Duration fit vs published	0.0766 d vs 0.0567 d (deviation 6.15σ)
Mid-time O–C	-1.3 min
Depth SNR	10.3
Residual scatter	1.43 %

Light curve

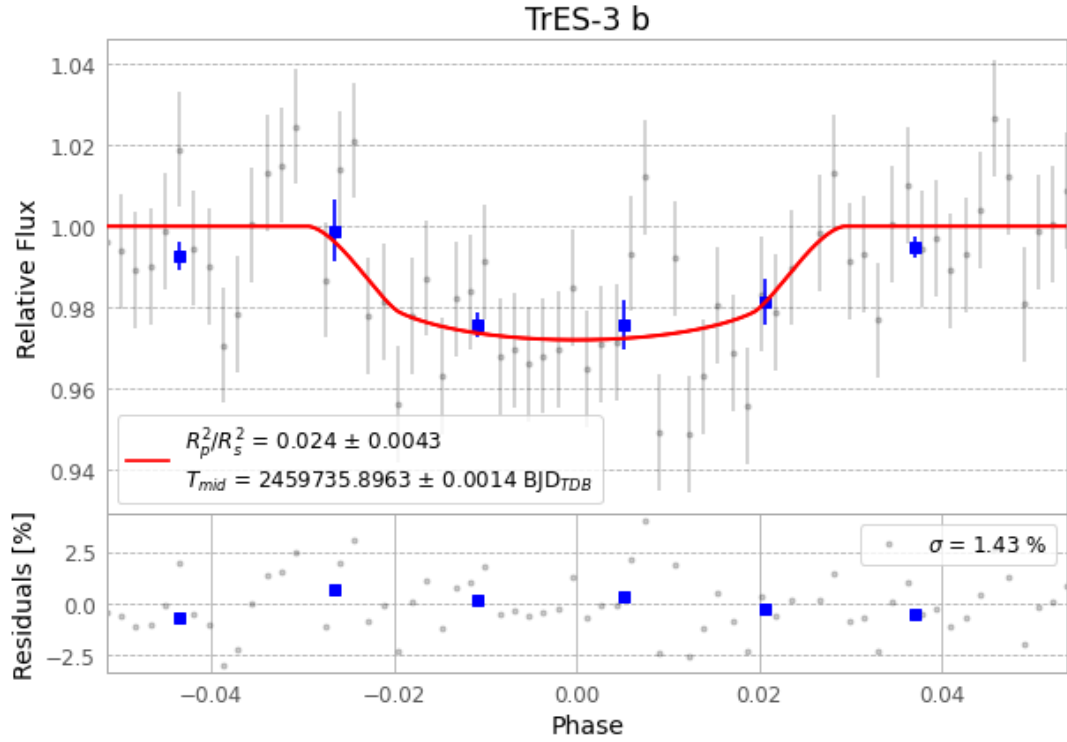


Figure 1 — FinalLightCurve_TrES-3 b_2022-06-05.png (SHA-256 30a6a0d9a18ad385...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [374, 226], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 73addbef1cfca108...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

68 FITS frames (45 MB), TRES-3220605074816.FITS → TRES-3220605110913.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-220605.log (c5f109c8ef31...), FinalLightCurve_TrES-3 b_2022-06-05.png (30a6a0d9a18a...), FinalLightCurve_TrES-3 b_2022-06-05.csv (ece3ce821aa7...)

Compute resources

Wall time 128.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 23:35Z.